

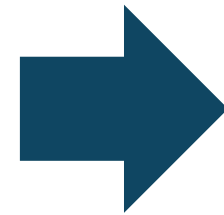
CPSC 4810 FINAL PROJECT

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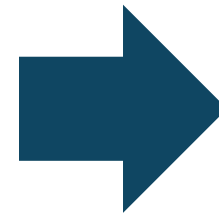
July 30, 2025

Customer Retention Strategy

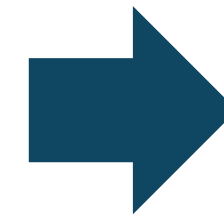
Defining Problem Statement



Data Collection



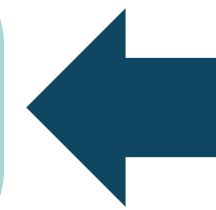
Data Cleaning & Transformation



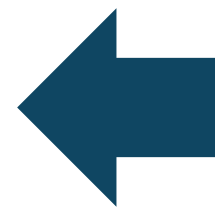
Exploratory Data Analysis (EDA)



Data Transformation & Feature Engineering



Machine Learning Model (Prediction/Clustering)



Dashboard



Key Insights & Business Recommendation



Defining Problem Statement

Market Context

Telecom market is highly saturated and there is virtually zero switching costs for customers hence telecom providers face high risks of customer churn. Meanwhile, the cost of acquiring new customers remains significantly higher than the cost of retaining existing ones.

Business Need

To remain competitive and profitable, telecom companies must prioritize customer retention. Therefore, identifying customers who are likely to churn and intervening before they do is not just desirable but essential.

Proposed Solution

Technical: Building a predictive model that identifies customers at risk of churning.

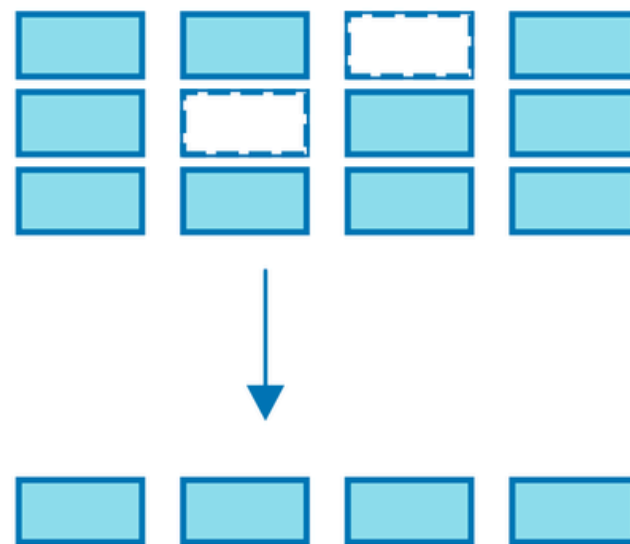
Business: Proactive retention strategies through actionable insights to improve customer engagement, and drive business value by reducing overall churn rates.

Data Collection

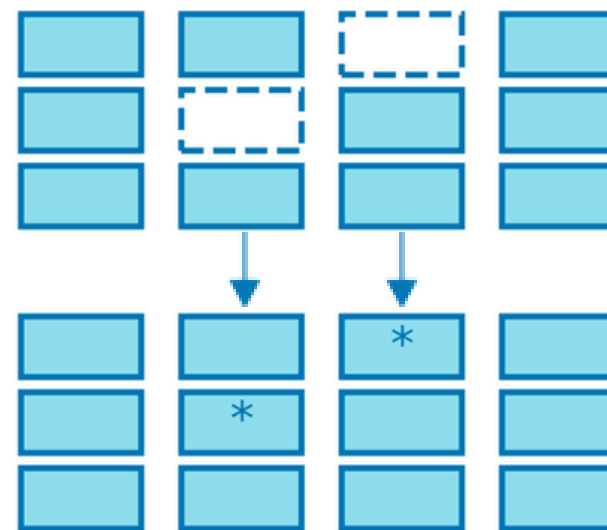
- **Source:**
 - Data was collected from Kaggle
 - <https://www.kaggle.com/jpacse/datasets-for-churn-telecom>
- **Dataset Description:**
 - This dataset provides customer-level data to predict churn in a telecom company.
 - It includes variables such as:
 - Customer demographics
 - Service usage metrics (e.g., Monthly Minutes, Total Recurring Charges)
 - Billing and payment information
 - Customer service interactions
 - Churn flag (Yes/No)

Data Cleaning & Transformation

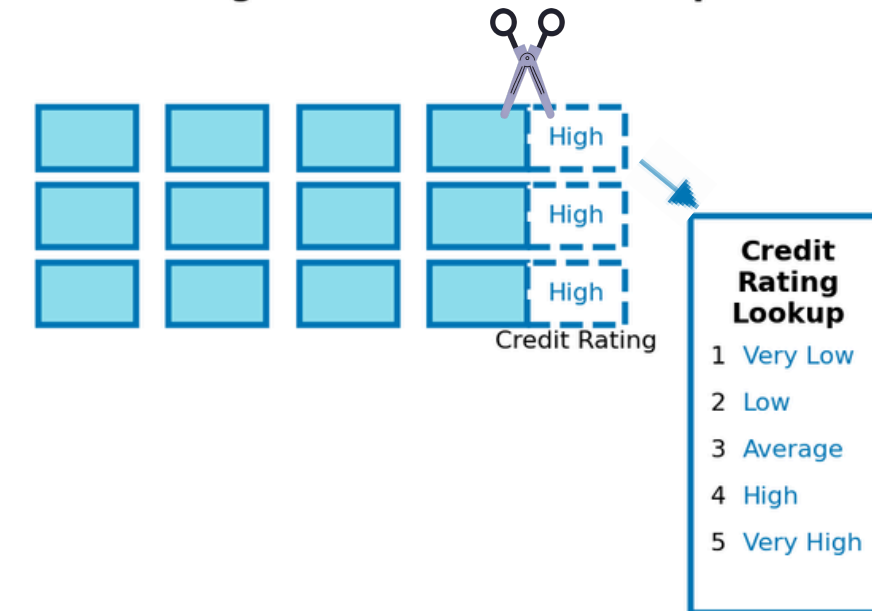
Missing Values Cleansing



Filling Missing Values



Move Categorical Column to Lookup Table



Exploratory Data Analysis

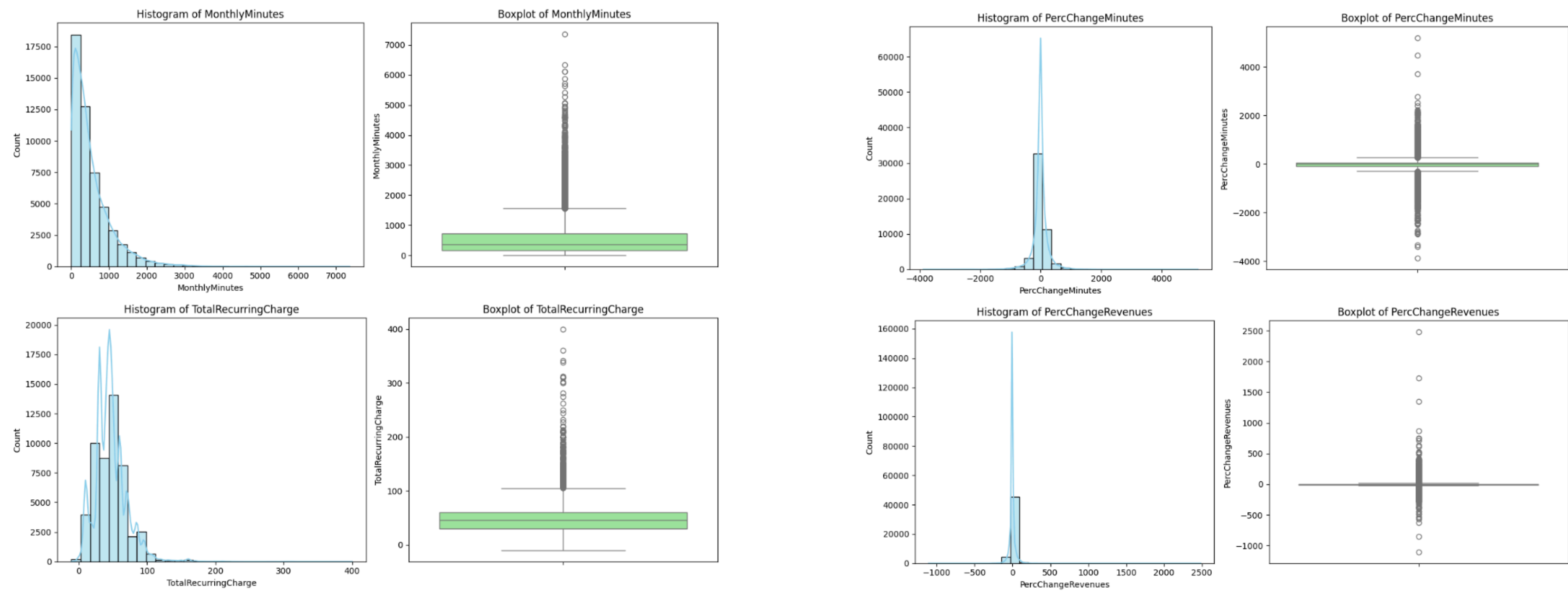
Descriptive Statistics

	MonthlyMinutes	TotalRecurringCharge	PercChangeMinutes	BlockedCalls	CustomerCareCalls	InboundCalls	CallWaitingCalls	MonthsInService	CurrentEquipmentDays
count	51047.00	51047.00	51047.00	51047.00	51047.00	51047.00	51047.00	51047.00	51047.00
mean	525.165514	46.824495	-11.464885	4.085672	1.868999	8.178104	1.840504	18.756264	380.544831
std	529.134170	23.812615	256.589244	10.946905	5.096138	16.665878	5.585129	9.800138	253.799599
min	0.00	-11.00	-3875.00	0.00	0.00	0.00	0.00	6.00	-5.00
25%	159.00	30.00	-82.00	0.00	0.00	0.00	0.00	11.00	205.00
50%	366.00	45.00	-4.00	1.00	0.00	2.00	0.30	16.00	329.00
75%	722.00	60.00	65.00	3.70	1.70	9.30	1.30	24.00	515.00
max	7359.00	400.00	5192.00	384.30	327.30	519.3	212.7	61.00	1812.00



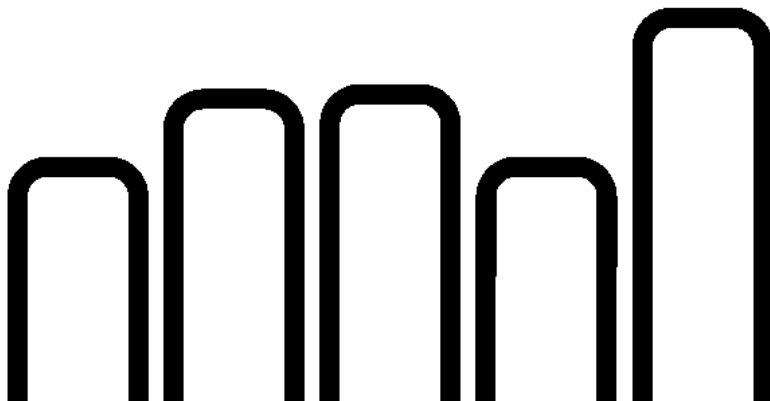
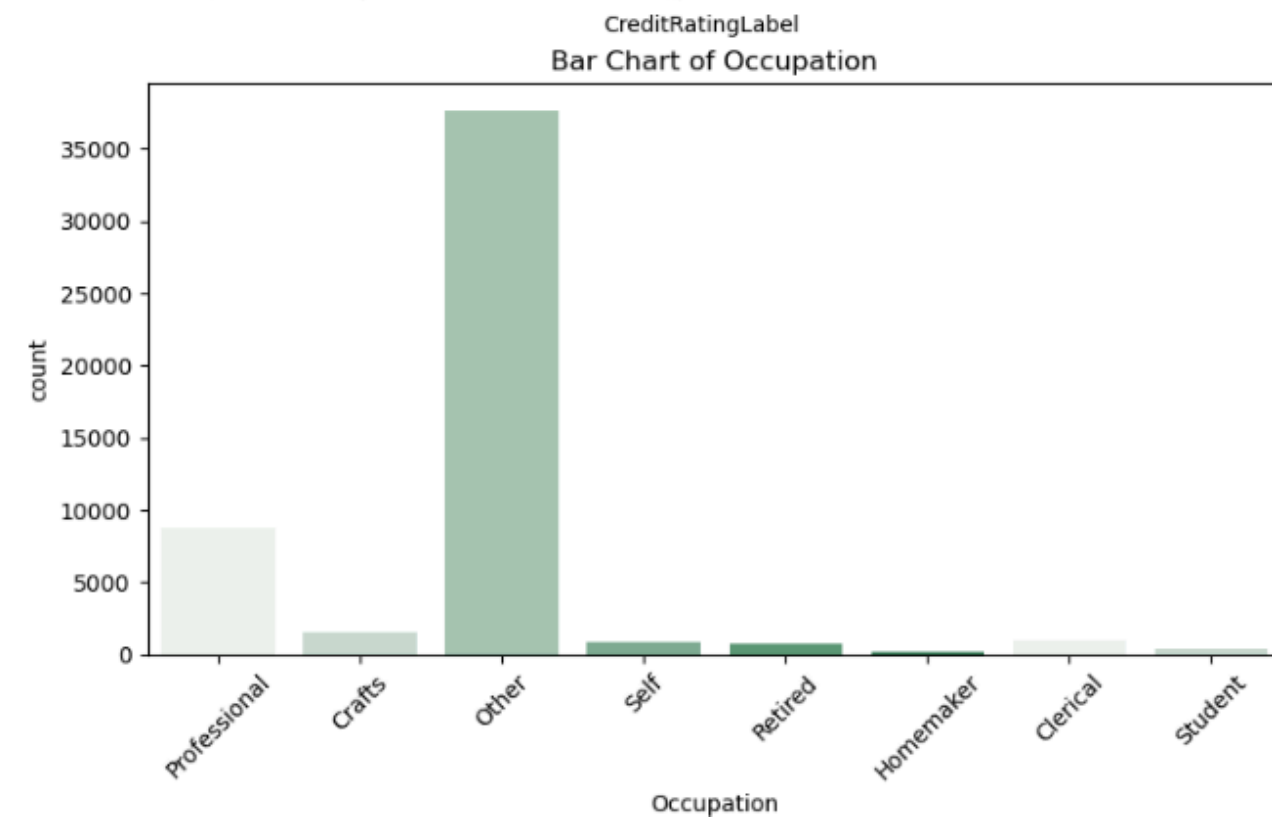
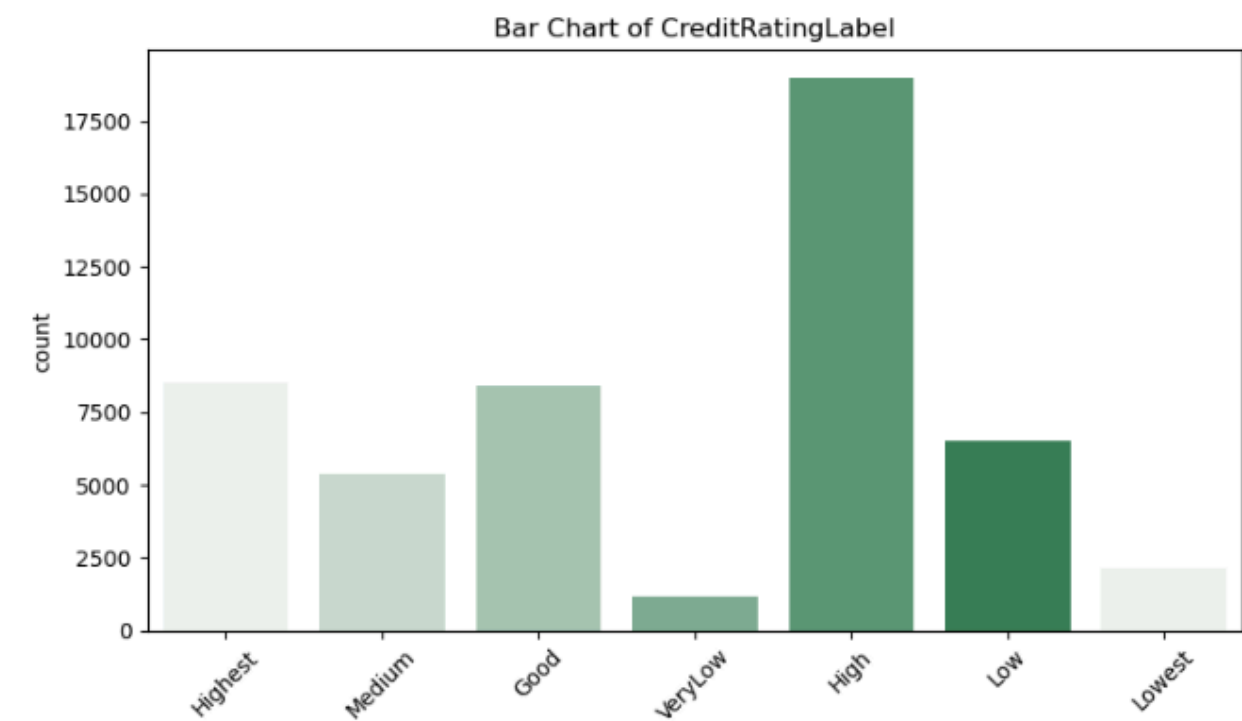
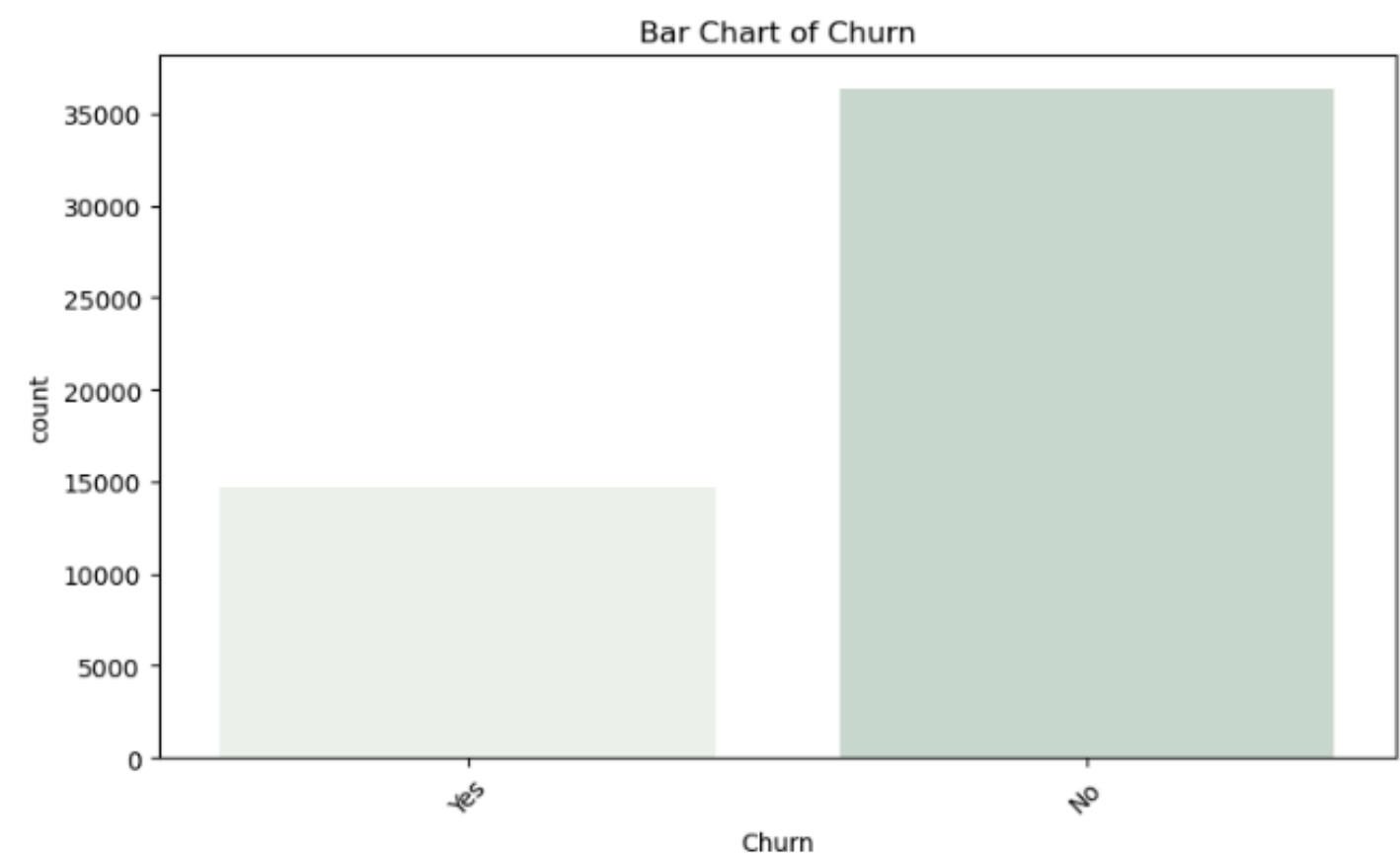
Exploratory Data Analysis

Visualizations



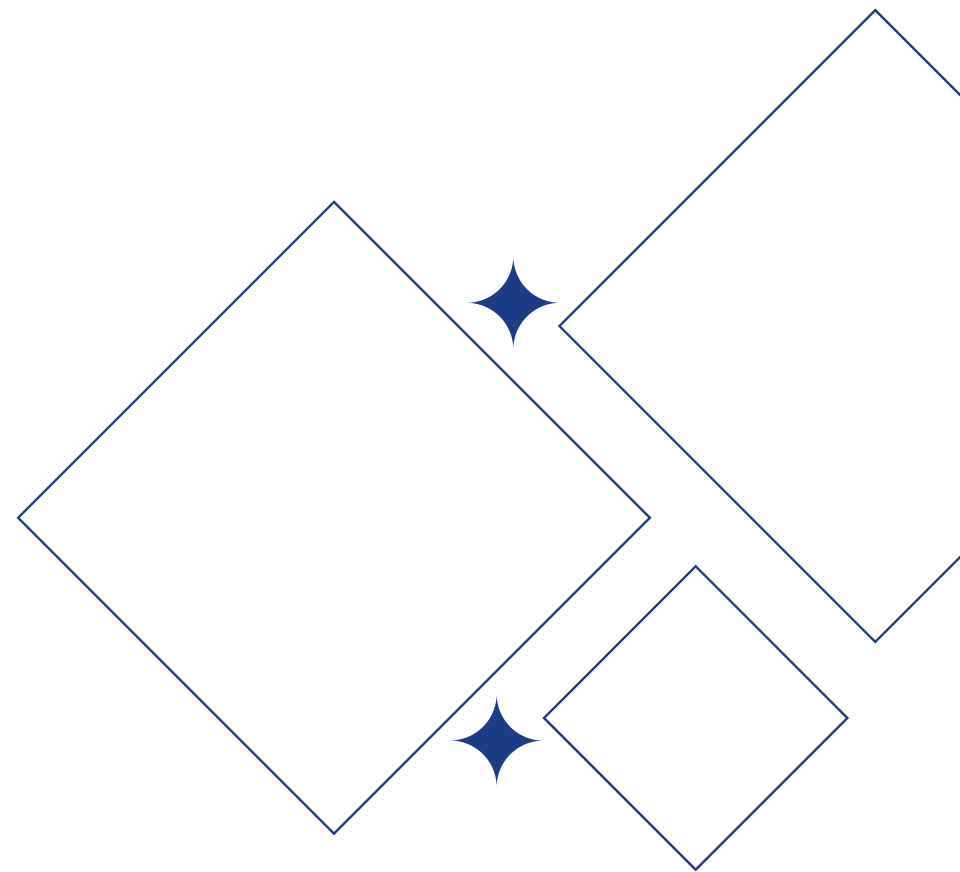
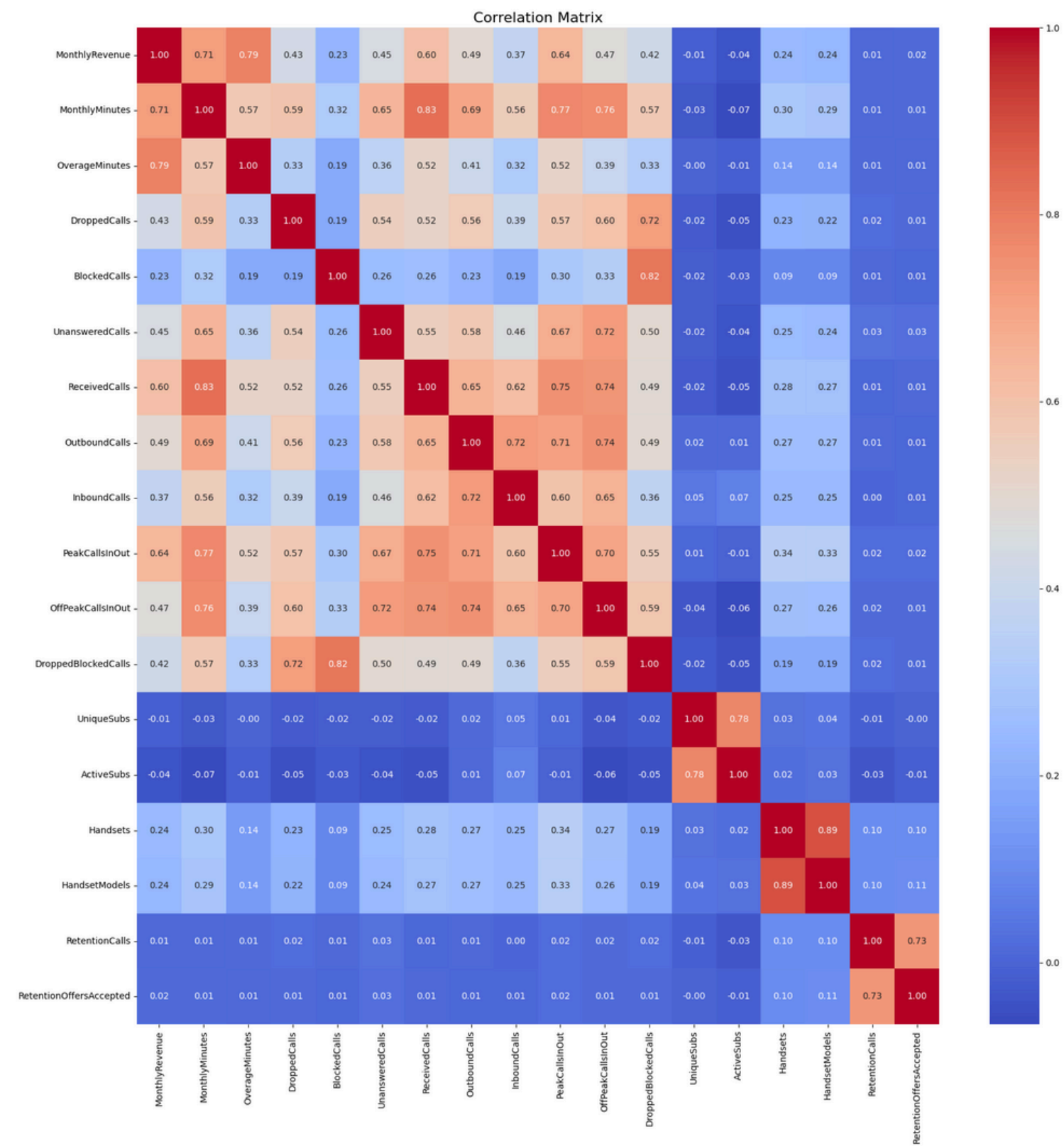
Exploratory Data Analysis

Visualizations



Exploratory Data Analysis

Visualizations



Featurering

STEP 1 – Encoding

- Binary
- Non binary
- One hot coding

STEP 2 – Address remaning missing value

- Drop variable with more than 30% missing value
- Ensure no missing value before proceed to feature selection

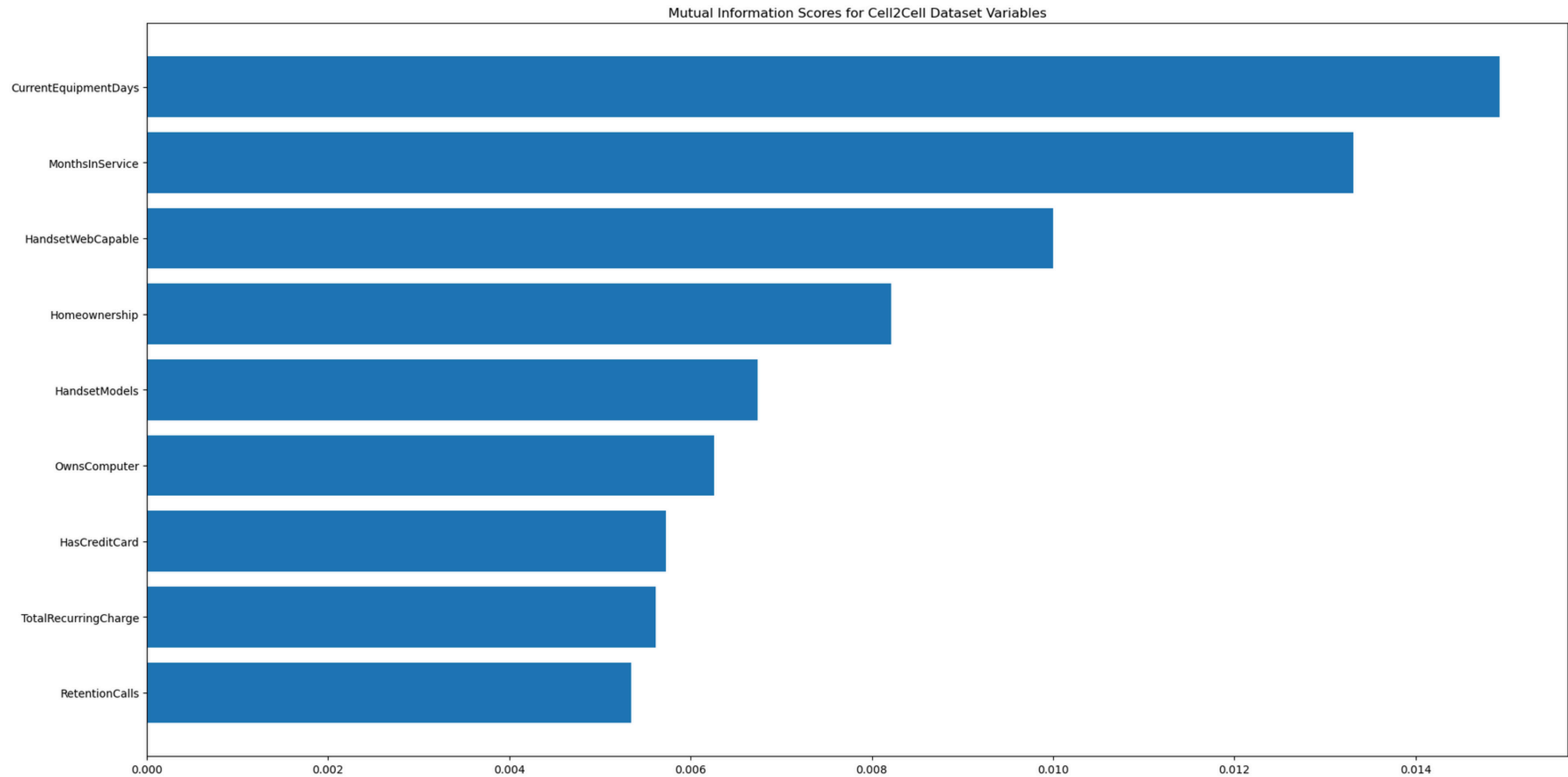
STEP 3 – Feature selection

- Dropped response variable and used 'mutual_info_classif' to select features that are valuable to the model

STEP 4 – Remove highly correlated feature

- Using threshold of 0.7, 19 variables were removed

Featurering



Featurering

Step 5

Normalising numerical variables to ensures all features contribute equally to the model

Other highlights

Principal Component Analysis were also performed, but no marginal value was added to the predictive model; hence, they were removed

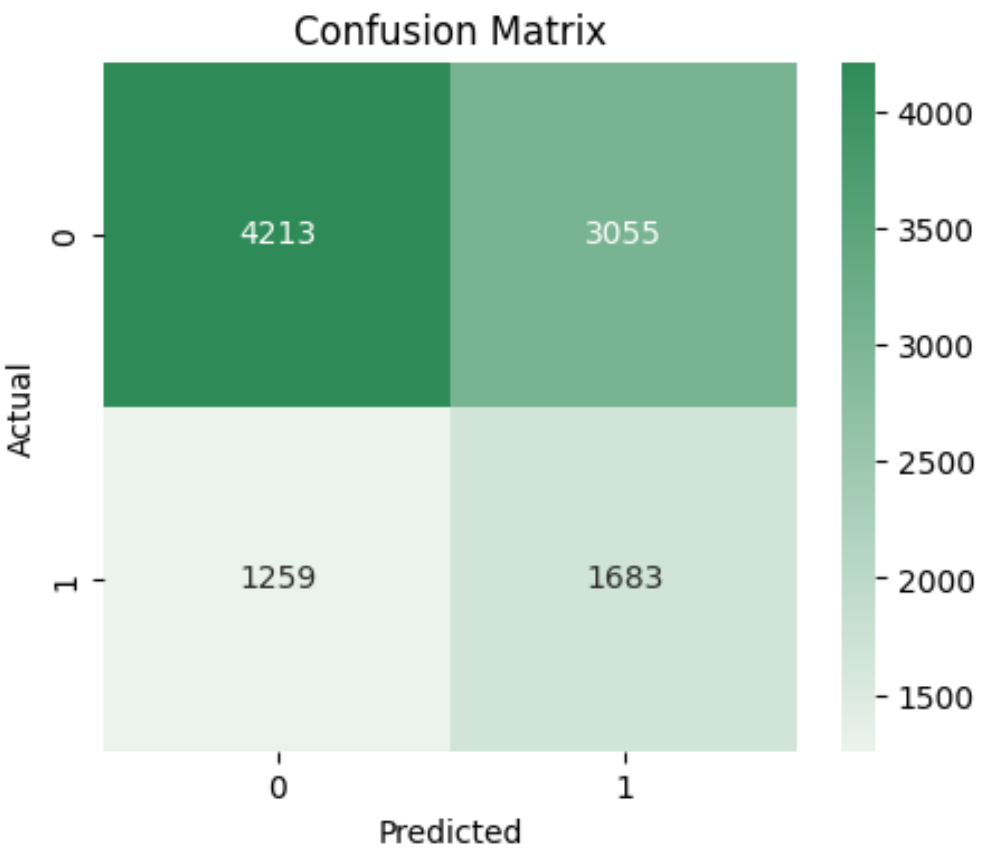
Conclusion

A total of 44 variables were retained for use in model prediction

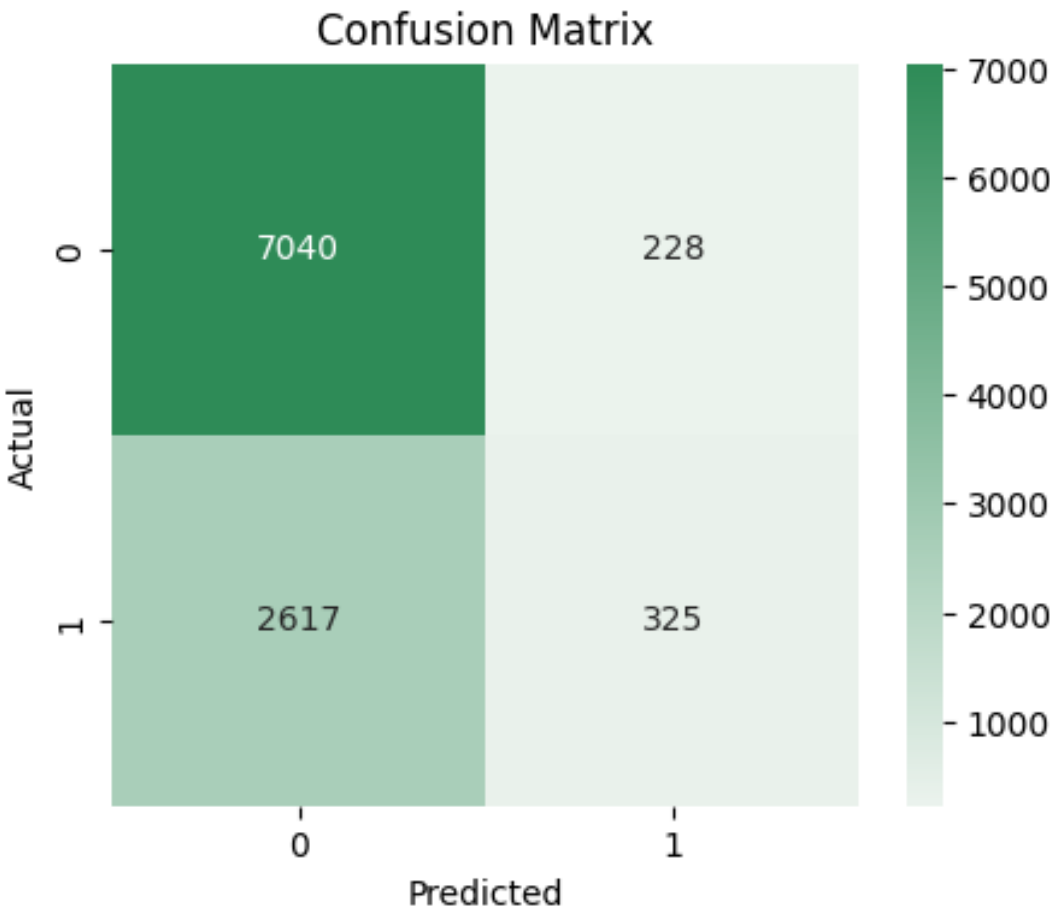
Modeling

Model	Classification	Precision	Recall	f1-score	Accuracy	Support
Logistic Regression	0	0.77	0.59	0.66	0.58	7268
	1	0.36	0.56	0.43		2942
XG Boost	0	0.73	0.97	0.83	0.72	7268
	1	0.6	0.1	0.18		2942

Logistic Regression



XGBoost

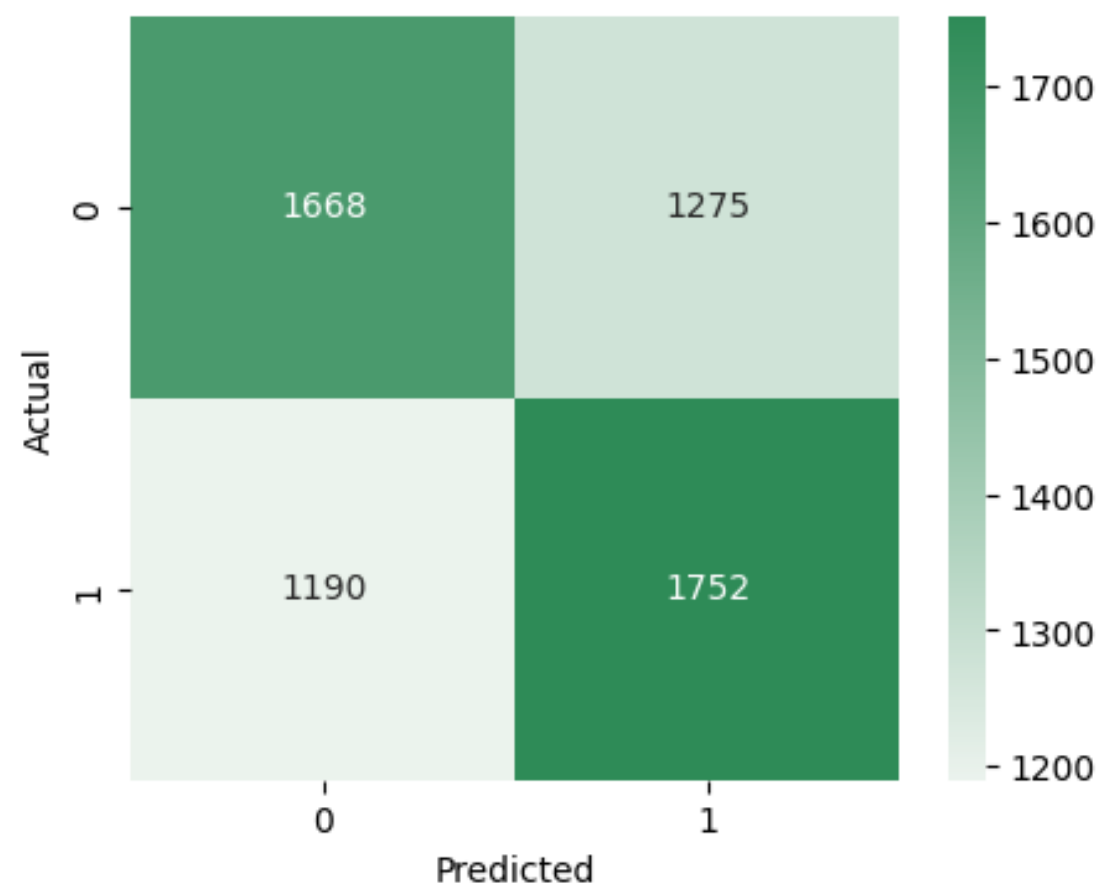


Modeling

Model	Classification	Precision	Recall	f1-score	Accuracy	Support
Logistic Regression Balanced	0	0.57	0.58	0.58	0.57	2943
	1	0.57	0.56	0.57		2943
XG Boost Balanced	0	0.63	0.59	0.61	0.62	2943
	1	0.61	0.65	0.65		2943

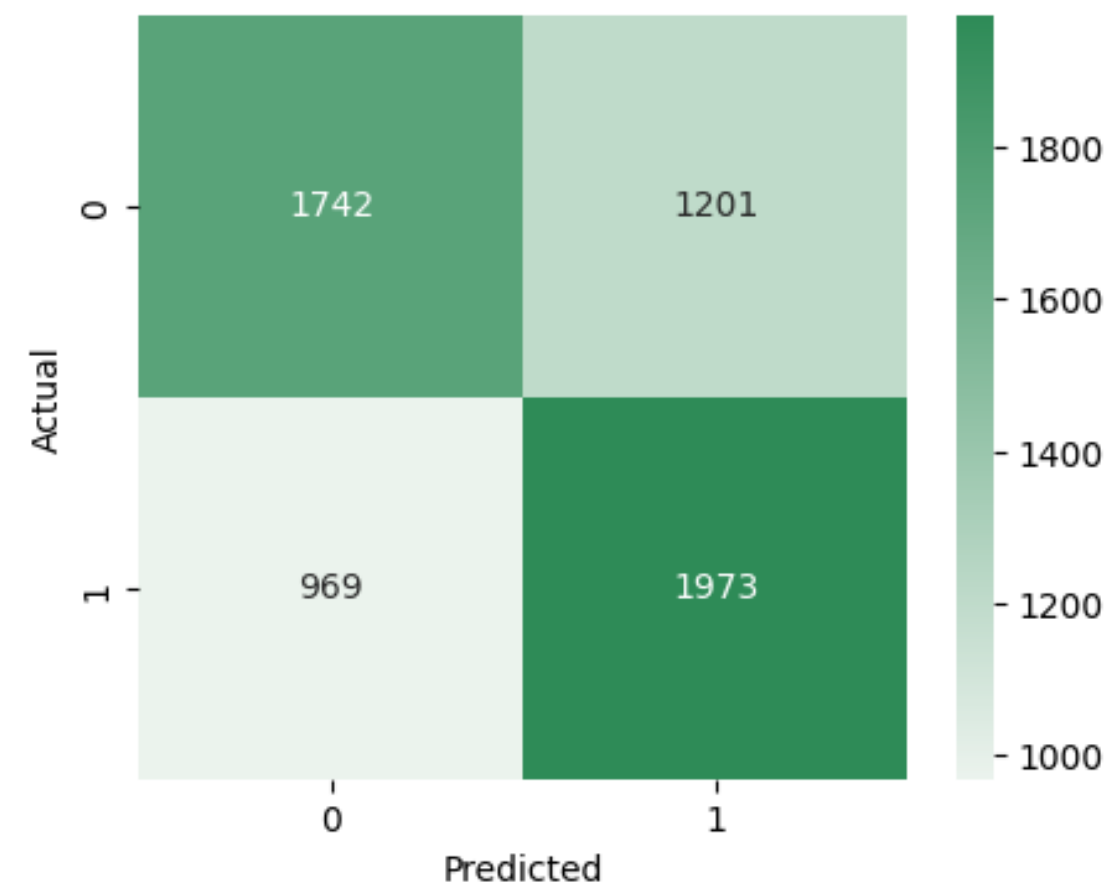
Logistic Regression Balanced

Confusion Matrix

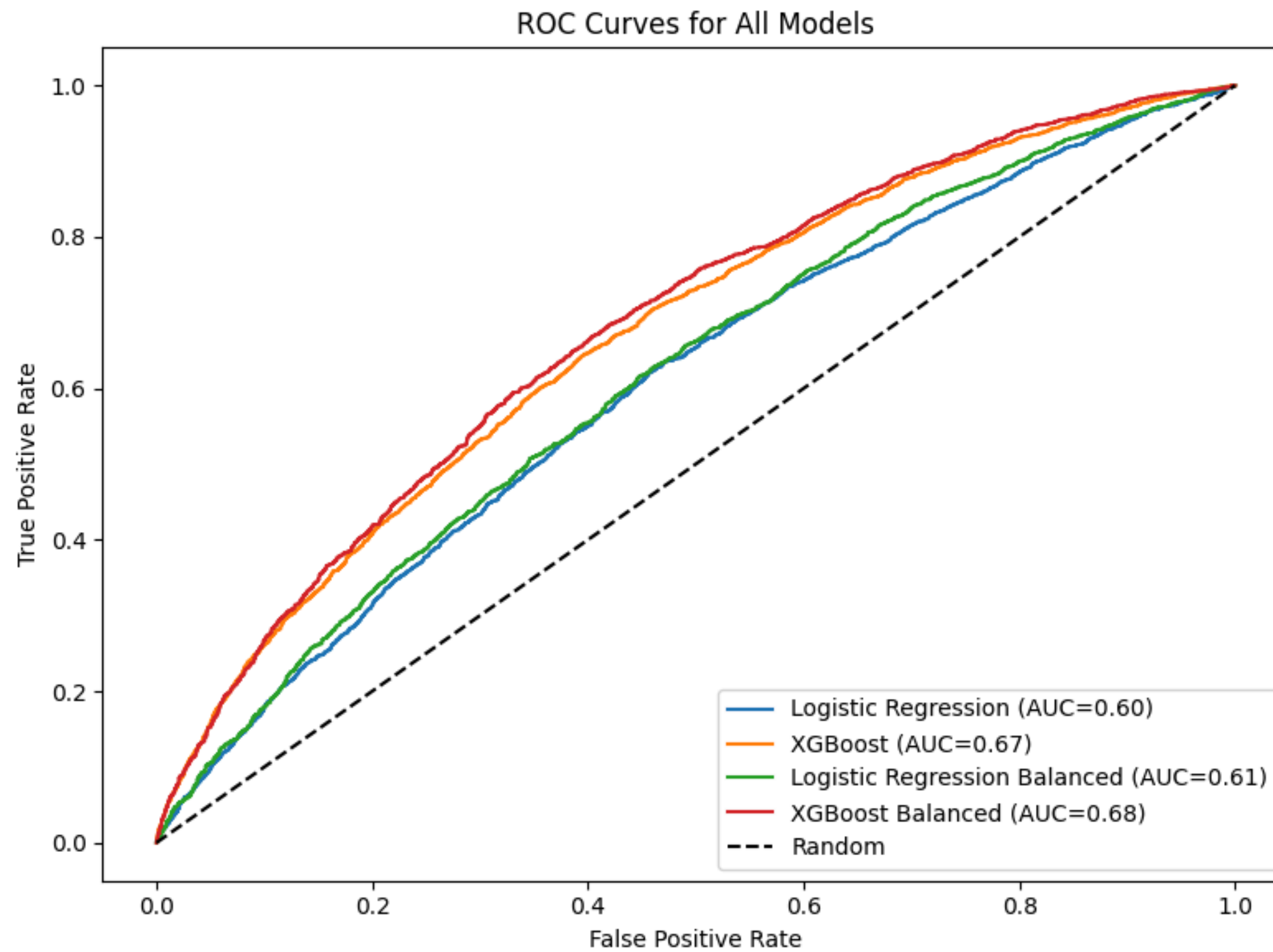


XGBoost Balanced

Confusion Matrix



ROC Curve



Dashboard

- **Python**
 - **Streamlit**
 - **Joblib**
 - **Plotly**
- **Demo**

Key insights & Recommendation

Predictive Modeling Shows Potential for Early Churn Detection

An accuracy of 0.68, indicating a reasonable starting point for identifying at-risk customers

Room for Model Improvement

- The model's performance suggests that current data may not fully capture the complexity of churn behavior.
- Lack of real-time data, limited socio-demographic variables, or absence of customer sentiment (e.g., from surveys or call transcripts) may be affecting predictive power

Key insights & Recommendation

Implement Targeted Retention Strategies Based on Current Model

We recommend launching a tiered retention program, starting with the top 20% highest-value customers

- Offering tailored incentives for the next billing
- Loyalty & Reward Programs: Introduce point-based or milestone reward systems to build long-term attachment
- Referral Programs: Offer rewards for bringing in friends or family — this strengthens personal investment in staying with your service

Expand Data Sources for Enhanced Predictive Power

Explore integrating customer feedback, sentiment analysis, social media interactions, and network quality indicators to enrich the feature set

Establish a Feedback Loop for Continuous Improvement

- Monitor post-intervention outcomes to validate the impact of the model-driven strategies.
- Use business KPIs (e.g., retention rate lift) to guide ongoing model tuning and strategy adjustments.

**Thank you
very much!**

PRESENTED BY STRONG TEAM